



PUNJAB UNIVERSITY COLLEGE OF INFORMATION TECHNOLOGY

Project Topic:

EVENT MANAGEMENT SYSTEM

Course Name:

Software Engineering

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Event Management System

Problem Statement

With the increasing number of events such as seminars, workshops, conferences, weddings, and social gatherings, managing event-related activities manually has become inefficient and error-prone. Event organizers often face difficulties in coordinating tasks, managing participants, and keeping accurate records.

The major challenges faced in event management include:

1. Event Display & Information Management

Event details such as event name, date, time, venue, and description are often scattered or maintained manually. This makes it difficult for participants to view complete and updated information about upcoming events.

2. Registration & Participant Handling

Managing participant registrations manually leads to errors such as duplicate entries, missing information, and difficulty in tracking registered attendees for each event.

3. Scheduling & Coordination Issues

Without a centralized system, scheduling conflicts may occur between events. Organizers also face challenges in coordinating event activities efficiently.

4. Event Preview & Planning Visualization

Organizers and clients often want to visualize event setups such as seating arrangements, themes, or schedules before finalizing the event. The lack of preview or planning visualization results in miscommunication and dissatisfaction.

5. Administrative & Monitoring Challenges

Tracking events, users, and daily activities manually makes it difficult for administrators to monitor system usage and generate reports for analysis and decision-making.

Proposed Solution

To overcome these challenges, we propose a **design-oriented Event Management System** that conceptually combines **interactive event display** with **administrative management capabilities** in a **single-machine software environment**.

The system is designed from a **Software Engineering perspective**, focusing on **requirement analysis, system behavior, and UML modeling**, rather than implementation or programming.

Reasoning for Using Single Machine Conceptual Software

1. Event Display & Interaction Mode

The system allows users to browse events interactively, view event details, and explore schedules. This mode is suitable for participants and organizers to understand event information clearly.

2. Management & Administration Mode

The same system supports administrative activities such as event creation, registration management, and report generation. The system seamlessly switches between user interaction and administrative management based on user roles.

This dual functionality ensures efficient event handling without the need for complex infrastructure.

Conceptual System Operation

The system operates on a **single local environment** and supports multiple user roles:

- **Admin** – Manages users, events, and reports
- **Organizer** – Creates and manages events
- **Attendee** – Views and registers for events

Based on the role, the system provides access to relevant functionalities.

Functional Breakdown of the Event Management System

1. Event Display & Information Module

- Displays upcoming and past events in a structured format
- Shows event details including date, time, venue, and description
- Supports searching and filtering of events

2. Registration Management Module

- Allows attendees to view event details before registration
- Manages participant registration records

- Prevents duplicate registrations
- Displays registration status for each event

3. Event Planning & Preview Module

This module helps organizers visualize and plan events effectively.

Key Features:

- **Event Setup Preview:**
Provides a conceptual preview of event plans such as schedules, sessions, or layouts.
- **Real-Time Updates:**
Any changes made to event details are reflected immediately in the preview.

This reduces planning errors and improves communication between organizers and stakeholders.

4. Admin Management Module

- Manages users and assigns roles
- Monitors overall system activities
- Generates reports such as number of events and registrations
- Tracks daily system usage

5. Daily Activity Tracking Module

- Tracks events created per day
- Tracks daily registrations
- Provides summary statistics for administrative review

Introduction

The **Event Management System (EMS)** is a conceptual software solution designed to streamline the planning, organization, and management of events such as seminars, workshops, conferences, weddings, and social gatherings. Traditional manual methods of managing events are often inefficient, error-prone, and time-consuming, leading to difficulties in tracking participants, maintaining accurate records, coordinating schedules, and generating reports.

Traditional Event Management (Before Improvements)

- Event management was handled manually using paper records, spreadsheets, or emails.

- Organizers had to perform multiple tasks individually:
 1. Record event details such as name, date, venue, and description
 2. Track participant registrations manually
 3. Coordinate schedules, resources, and venues without centralized oversight
 4. Generate reports and summaries manually
- This process was prone to errors, caused delays, and made communication with participants cumbersome.

Event Management System (After Improvements)

- The improved EMS provides a **centralized, design-oriented solution** that handles all event-related activities efficiently.
- Key functionalities include:
 1. Creating, updating, and deleting events with complete details
 2. Allowing participants to view event information and register seamlessly
 3. Monitoring daily activities and generating administrative reports
 4. Assigning roles (Admin, Organizer, Attendee) to ensure secure, role-based access
- The system offers a clear, interactive overview of events, participant lists, and schedules, improving both accuracy and workflow efficiency.

Key Enhancements

- Centralized and structured management of events and registrations
- Role-based access control for different user types
- Quick and accurate generation of reports and daily activity summaries
- Reduced errors, improved coordination, and streamlined event workflows
- Design-oriented, academic-focused approach that provides a conceptual blueprint for future system implementation

USECASE DOCUMENTATION

UC ID: EMS-101-347

Title: Event Management system

Primary Actor: Attendee / Participant

Secondary Actor: System, Payment Gateway, Email Service (for notifications), Admin (for approvals/support)

Preconditions

1. Participant has access to the Event Management System (web or desktop).
2. Participant has a valid account or is ready to register.
3. System is operational and online.
4. Participant has access to a payment method (if the event requires a registration fee).

Main Flow

Success Scenario

1. Participant navigates to the Event Management System.
2. Participant selects **Login**.
3. System displays login form.
4. Participant enters valid credentials (email/username and password).
5. System authenticates the user and displays the participant dashboard/homepage.
6. Participant selects **View Events**.
7. System displays a list of upcoming events with basic details (name, date, venue, registration fee if any).
8. Participant selects a specific event to view more details.
9. System displays detailed event information: schedule, sessions/workshops, venue, available seats, description, and registration fee.
10. Participant clicks **Register for Event**.
11. System displays the registration form with required fields: name, email, contact number, and optional preferences.
12. Participant fills in all required information.
13. System validates the input.
14. Participant submits the registration form.

15. If the event requires a fee:
 - 15.1 System displays the payment options (credit card, debit card, e-wallet, etc.).
 - 15.2 Participant selects a payment method and enters payment details.
 - 15.3 System processes the payment through the payment gateway.
 - 15.4 System confirms successful payment.
16. System displays a registration summary for participant confirmation.
17. Participant reviews and confirms the registration.
18. System allocates a unique registration ID and updates the participant list.
19. System optionally sends a confirmation email with registration details and payment receipt.
20. System displays a final confirmation message with registration ID, event details, and payment status.

Main Flow – Failure Scenarios & Alternatives

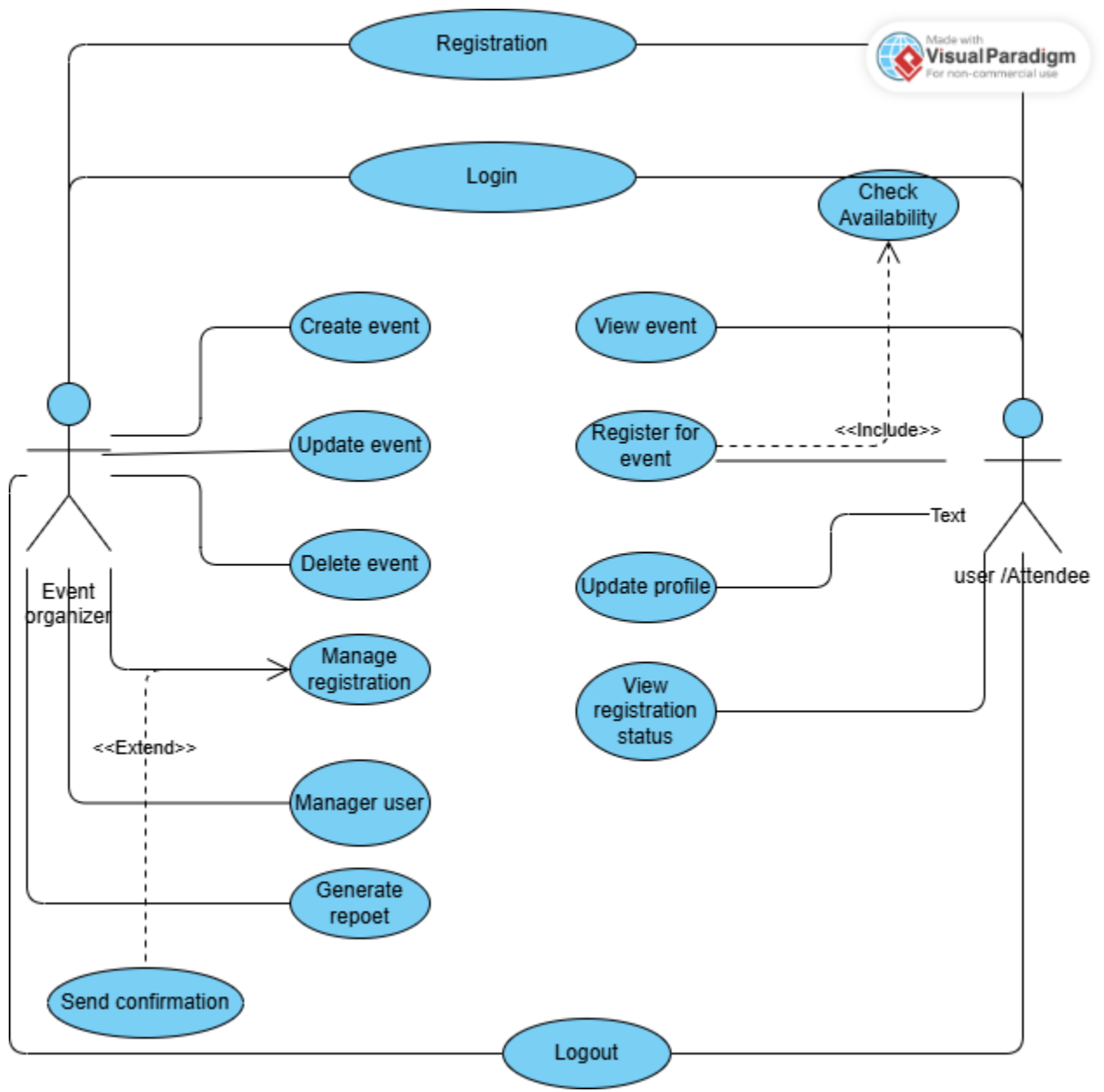
1. **Invalid Login Credentials**
 - 1.1 Participant enters incorrect email or password.
 - 1.2 System displays “Invalid email or password.”
 - 1.3 Participant retries login.
 - 1.4 After 3 failed attempts, system locks account temporarily and displays password reset instructions.
2. **Event is Fully Booked**
 - 2.1 Participant selects an event with no available seats.
 - 2.2 System displays “Event is fully booked.”
 - 2.3 Participant is offered to join the waitlist.
 - 2.4 Participant confirms waitlist registration.
 - 2.5 System updates the waitlist and sends confirmation.
3. **Participant Selects Multiple Sessions & One Session is Full**
 - 3.1 Participant chooses multiple sessions.
 - 3.2 System detects one or more sessions are full.
 - 3.3 System displays alert and suggests alternative sessions.
 - 3.4 Participant selects available sessions.
4. **Invalid Participant Information**
 - 4.1 System detects missing or invalid mandatory fields.

- 4.2 System highlights fields and prompts correction.
- 4.3 Participant corrects the input and resubmits.
- 4.4 If repeated failures (>3 attempts) occur, system prompts admin intervention.
- 5. **Event Expired or Cancelled**
 - 5.1 Participant selects an event that is no longer available.
 - 5.2 System displays "Event not available."
 - 5.3 System shows alternative upcoming events.
 - 5.4 Participant selects a new event.
- 6. **Registration/Confirmation Failure (Network or Internal Error)**
 - 6.1 System fails to complete registration.
 - 6.2 System displays "Registration could not be completed. Please try again."
 - 6.3 Participant retries registration.
 - 6.4 If failure persists, system logs error and notifies admin.
- 7. **Payment Failure**
 - 7.1 Payment gateway fails or transaction is declined.
 - 7.2 System displays "Payment unsuccessful. Please try again or use a different payment method."
 - 7.3 Participant retries payment or selects alternative method.
 - 7.4 If repeated failures occur, participant registration is temporarily held until payment is successful.
- 8. **Exceeding Registration Limits**
 - 8.1 Participant attempts to register for more events than allowed.
 - 8.2 System displays warning and suggests reducing selected events or splitting registration across accounts.
- 9. **Email Confirmation Fails**
 - 9.1 System fails to send confirmation email.
 - 9.2 System displays "Email failed. Download receipt directly."
 - 9.3 System logs error for admin review.

Postconditions

- 1. Participant is successfully registered for the selected event.
- 2. Payment (if required) is successfully processed.
- 3. Participant details are updated in the system.
- 4. Confirmation (ticket/receipt/registration ID and payment receipt) is displayed and optionally emailed

USE CASE DIAGRAM



DOMAIN MODEL

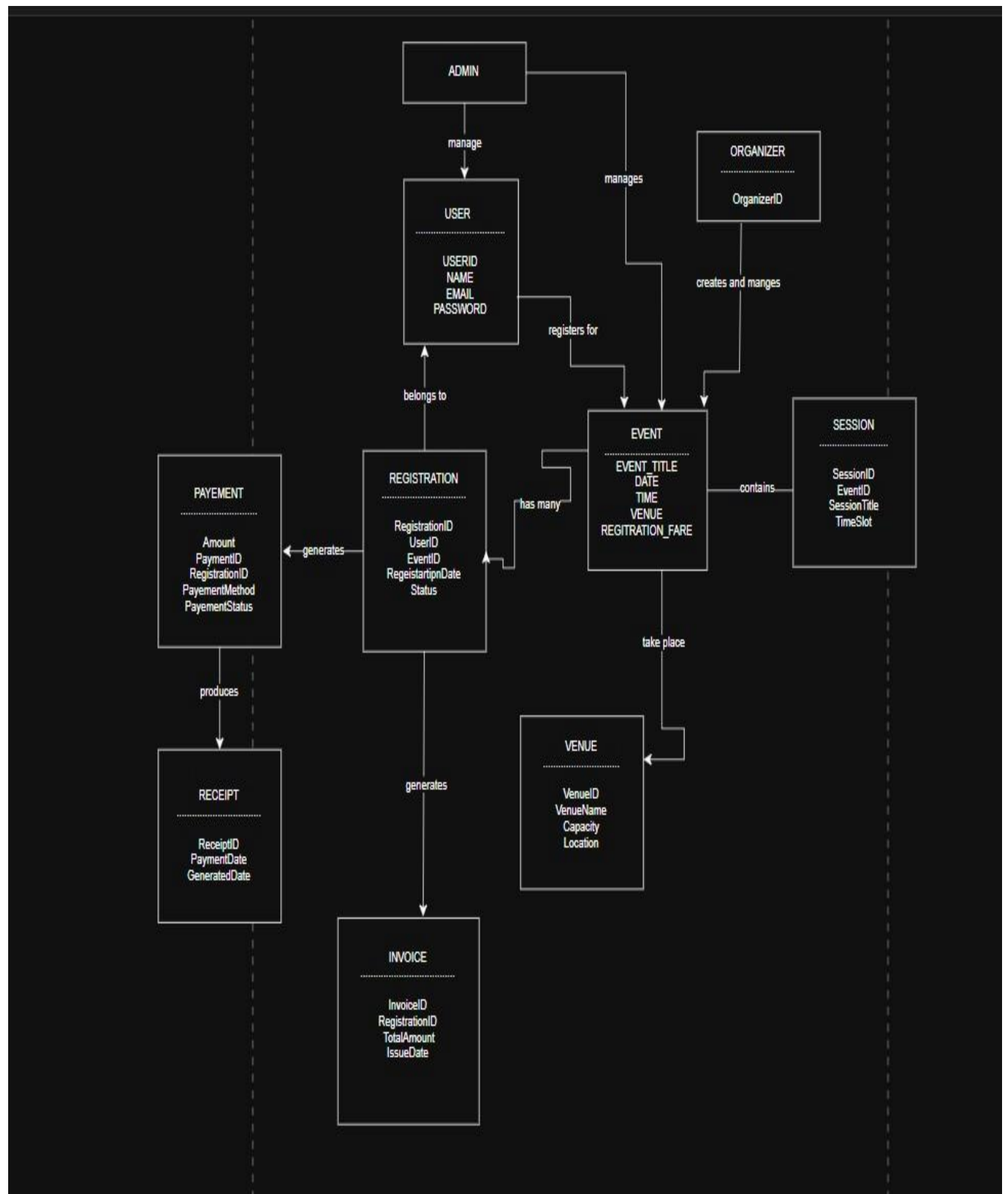
The Event Management System starts when a user logs into the application and is welcomed by a dashboard showing a variety of upcoming events. Each event is presented

with key information such as title, date, venue, organizer details, and registration fee. The user selects an event to view its complete description, schedule, available sessions, and remaining seats. After reviewing the details, the user proceeds to register by entering personal information and confirming participation. For paid events, the system generates a payment request and guides the user through the fee submission process. Once the payment is successfully completed, the system creates a registration record, assigns a unique registration ID, and updates the event's participant list. A confirmation message and receipt are then generated and sent to the user. Meanwhile, organizers and administrators use the system to manage events, monitor registrations, and ensure that all event activities are properly coordinated and recorded.

ENTITIES

Candidate Entities	Discarded Entities
User	System
Admin	Login Screen
Organizer	Button
Attendee	Message
Event	Event Name
Registration	Date
Payment	Amount
Invoice	Text
Receipt	Notification
Session	Description
Venue	Location
Report	Data

DOMAIN MODEL DIAGRAM



ACTIVITY DIAGRAM

